

Clifford Algebra

Sagnik Seth

**Dept of Physical Sciences
IISER Kolkata**

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1 Algebra

My introduction to Clifford Algebra was through a very confusing class on Dirac equation in Quantum Field Theory. I wanted to understand exactly all the fuss about spinors and Clifford Algebra were and for that I found that I had to learn a hell lot of rigorous math. It might take me decades to understand this fully, however, I thought to atleast start and see where it goes. The obvious starting point is to define what an *algebra* is and I too shall try to start at that.

Many times, we have seen that a particular vector space is occupied with some ‘multiplication’ operations, say like, the space of matrices is endowed with matrix multiplication. Here we will try to characterise such spaces with a ‘product’ and see what properties these spaces have.

Definition 1 (Algebra):

An *algebra* $\mathcal{A}$ over a field $\mathbb{F}$ is a vector space with a bilinear binary operation $\cdot : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$. Thus for all $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathcal{A}$ and $\beta, \gamma \in \mathbb{F}$ we have,

$$\begin{aligned}\mathbf{a} \cdot (\beta \mathbf{b} + \gamma \mathbf{c}) &= \beta \mathbf{a} \cdot \mathbf{b} + \gamma \mathbf{a} \cdot \mathbf{c} \\ (\beta \mathbf{b} + \gamma \mathbf{c}) \cdot \mathbf{a} &= \beta \mathbf{b} \cdot \mathbf{a} + \gamma \mathbf{c} \cdot \mathbf{a}\end{aligned}$$

Thus, from the definition atleast, we see that an algebra need not be associative. We call an algebra *associative* if $\mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c}) = (\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{c}$.

If there exists an element $\mathbf{e}$ such that $\mathbf{a} \cdot \mathbf{e} = \mathbf{e} \cdot \mathbf{a} = \mathbf{a} \forall \mathbf{a}$, then $\mathbf{e}$ is called the *identity element* or *unit* and the algebra is called a *unital algebra*.

Note that the identity of an algebra is unique. To see that, let us assume $\exists \mathbf{e}, \mathbf{e}'$. Since $\mathbf{e}$ and $\mathbf{e}'$ both are identities, this would imply that

$$\mathbf{e} \cdot \mathbf{e}' = \mathbf{e} \quad \mathbf{e} \cdot \mathbf{e}' = \mathbf{e}'$$

from which we can conclude that $\mathbf{e} = \mathbf{e}'$, hence the identity of an algebra is unique.

Definition 2 (Sub-Algebra):

Let $\mathcal{A}$ be an algebra and $\mathcal{A}'$ be a subspace of $\mathcal{A}$ which is closed under multiplication, that is, $\forall \mathbf{a}, \mathbf{b} \in \mathcal{A}', \mathbf{a} \cdot \mathbf{b} \in \mathcal{A}'$. Then $\mathcal{A}'$ is called a *subalgebra* of $\mathcal{A}$.

2 Clifford Algebra

Let us now start with the definition of the Clifford Algebra. Well, there are different ways of defining the Clifford Algebra, through a pure algebraic approach, through the exterior algebra, through tensor algebra, etc. All of these are mostly equivalent, however the definition of the Clifford Algebra as a quotient algebra of the tensor algebra generalises the notion to rings also, in addition to field.

I understand that all the hoity-toity things that I said above does not make any sense. Thus, we will try to forsake the rigour and understand Clifford algebra as intuitively as possible. The foremost thing to define a Clifford algebra is a vector space V over a field K , equipped with a quadratic form $Q : V \rightarrow K$.

For now, think of quadratic forms as a polynomial expression with terms of degree 2 (eg. $x^2, y^2, xy \dots$). What Q does it, it takes some $\mathbf{v} \in V$ and then represents $\mathbf{v}$ in some basis $\mathcal{B}$ of V and then puts the coordinates of $\mathbf{v}$ in that basis into the polynomial expression to generate some element in the field K .

Here we will consider Clifford algebras as an extension of the *exterior algebra* which was introduced by Grassman.

2.1 Exterior Algebra

This is the algebra introduced by *Grassmann* where the bilinear binary operation of the algebra is the *wedge product* or the *exterior product*. The exterior product of vectors comes from differential geometry and a basic introduction can be found [here](#)!